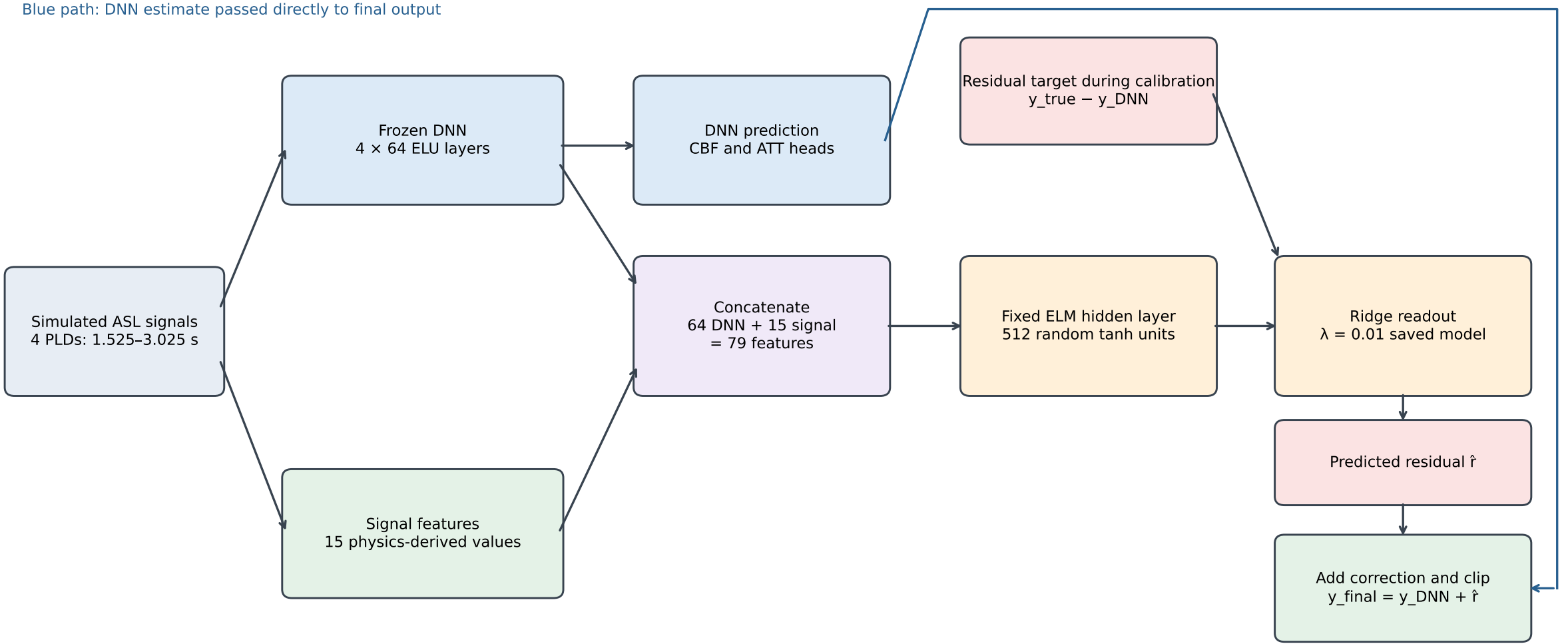
Implemented H-DELM inference and residual-calibration flow
Blue path: DNN estimate passed directly to final output
Simulated ASL signals
4 PLDs: 1.525–3.025 s
Frozen DNN
4 × 64 ELU layers
DNN prediction
CBF and ATT heads
Signal features
15 physics-derived values
Concatenate
64 DNN + 15 signal
= 79 features
Fixed ELM hidden layer
512 random tanh units
Residual target during calibration
y_true − y_DNN
Ridge readout
λ = 0.01 saved model
Predicted residual r̂
Add correction and clip
y_final = y_DNN + r̂